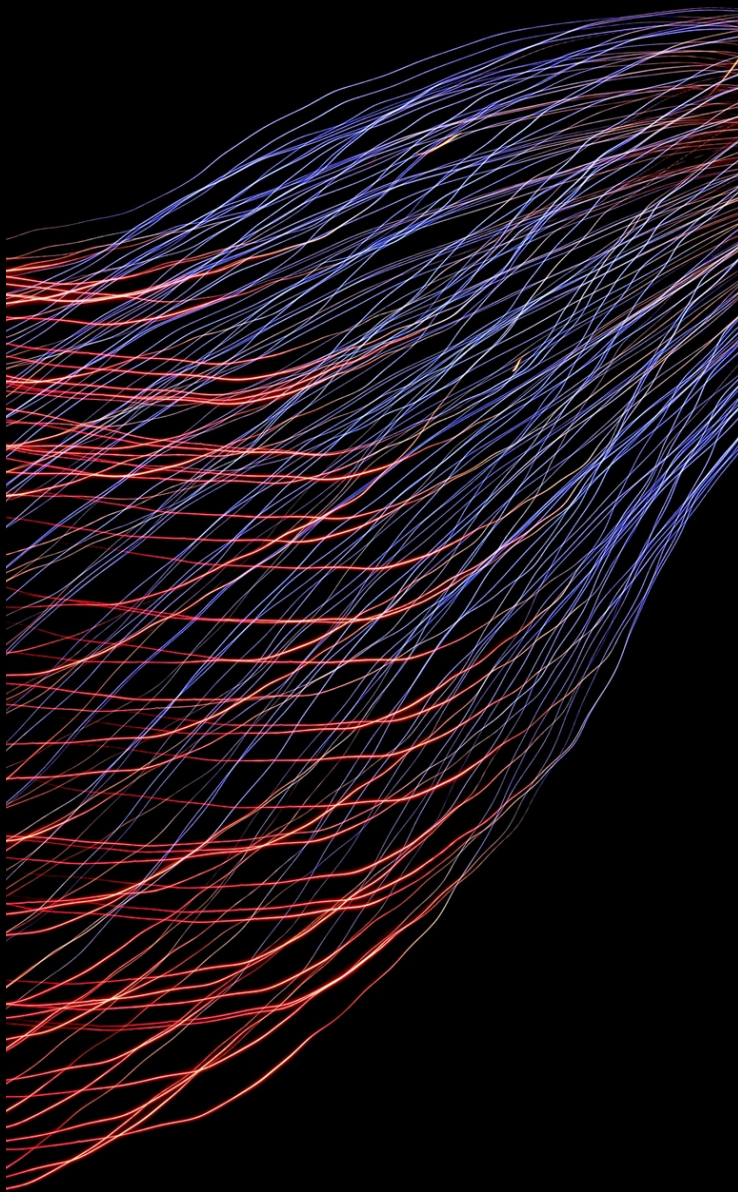




Model Validation and Evaluation: A Deep Dive into AUC-ROC and Cross Validation

Brief overview of model validation strategies like AUC-ROC and cross-validation.



AUC - ROC

Area Under the Receiver Operating Characteristic Curve

It is a performance metric used to evaluate the quality of binary classification models.

Cross Validation

Fight Overfitting

Cross Validation is a resampling technique used to assess the generalizability of a model. It helps to mitigate overfitting by dividing the dataset into subsets and training the model multiple times on different combinations of these subsets.

What is Model Validation?

Definition

Model validation is the process of evaluating a model's performance and suitability for its intended purpose.

Key Goals

The key goals of model validation are to identify model limitations, quantify uncertainties, and ensure regulatory compliance.

Methods

Common methods for model validation include statistical tests, benchmarking, sensitivity analysis, and out-of-sample testing.

When to Validate

Models should be validated before deployment and periodically after deployment to ensure they remain sound over time.

Who Performs It

Model validation is often performed by a team independent from the model builders to avoid conflicts of interest.

Regulatory Requirements

Many regulations like Basel III require rigorous validation for models used in areas like credit risk and capital calculations.

How Does AUC-ROC Work?

TPR, FPR as Confusion Matrix

The ROC curve is plotted with the True Positive Rate (TPR) on the y-axis and the False Positive Rate (FPR) on the x-axis.

ROC Curve Plotting

The curve is generated by varying the classification threshold and calculating the TPR and FPR at each point.

AUC Interpretation

The AUC is then calculated as the area under this curve. An AUC value close to 1 indicates a good model, while a value close to 0.5 suggests no discrimination capability.

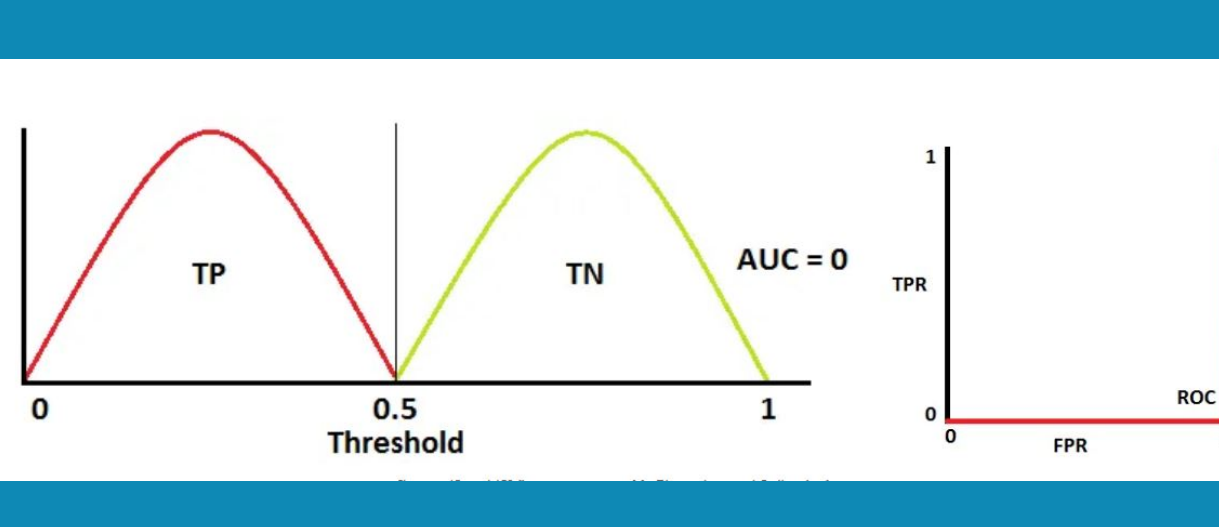
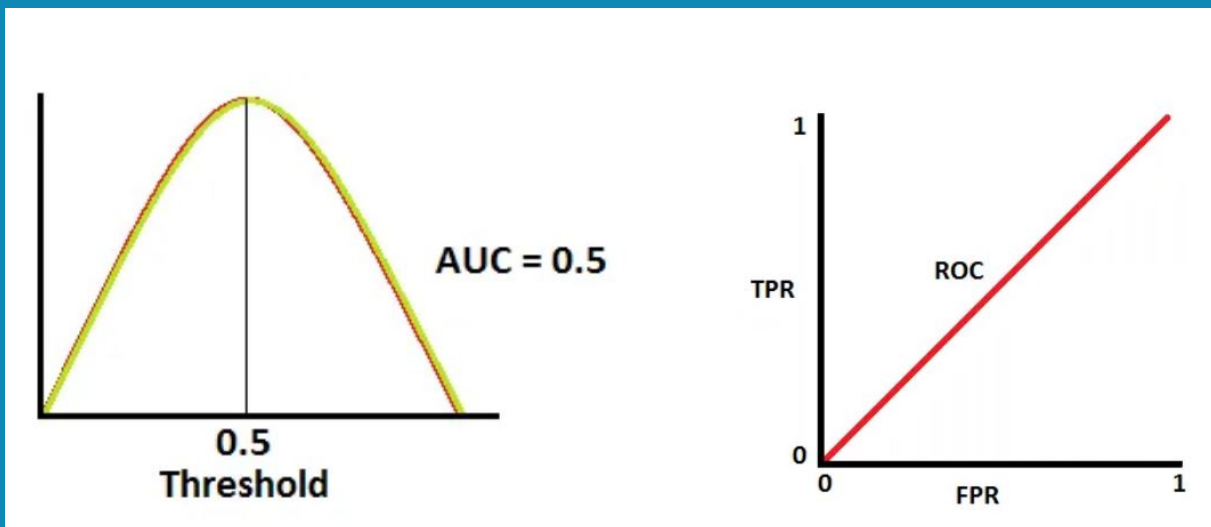
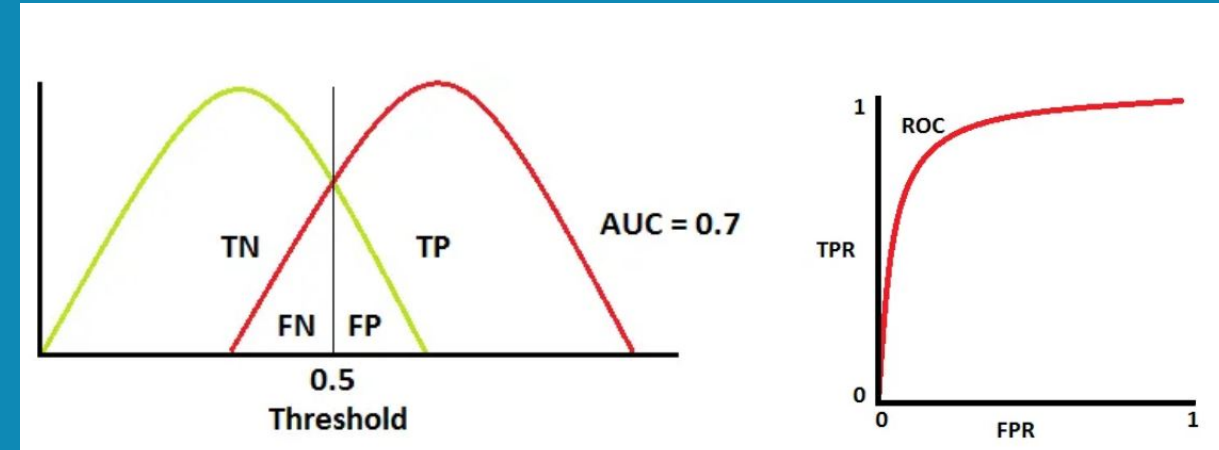
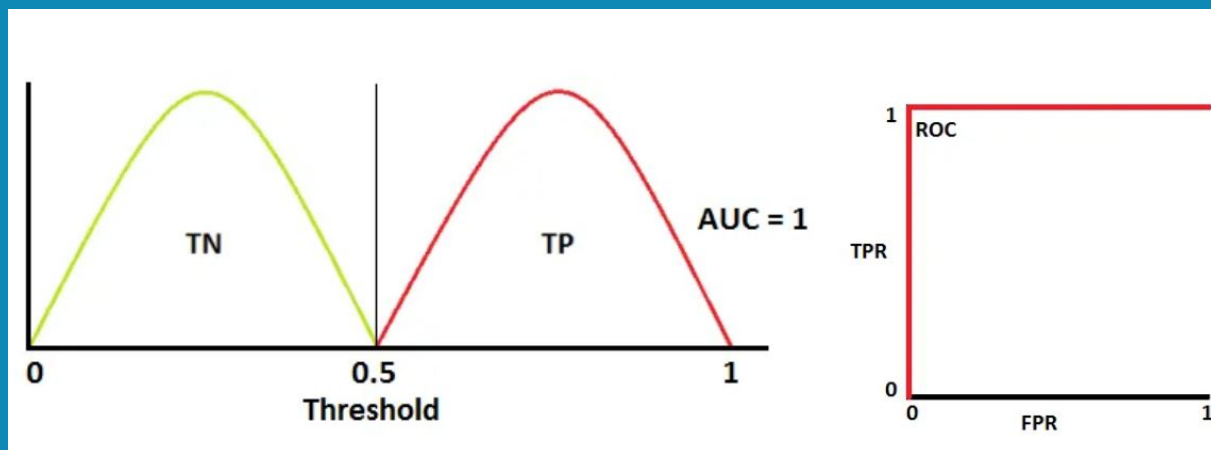
AUC-ROC in Credit Models

The AUC-ROC curve is a useful tool for evaluating classifier models in credit risk modeling. It provides a metric for comparing true positive and false positive rates across different classification thresholds.

$$TPR/Recall/Sensitivity = \frac{TP}{TP + FN}$$

$$Specificity = \frac{TN}{TN + FP}$$

$$FPR = 1 - Specificity = \frac{FP}{TN + FP}$$



Section 2

Cross - Validation

What is Cross Validation?

Definition

Cross validation is a resampling procedure used to evaluate machine learning models on a limited data sample.

K-fold Cross Validation

The data is split into k groups, or folds, of equal size. The model is trained using $k-1$ folds and validated on the remaining fold.

Repeated Cross Validation

Repeating cross validation multiple times using different splits helps reduce variability and provides more reliable estimates.

How Does Cross Validation Work?

- **Split data into k folds**

Split the dataset into k equal sized folds or parts

- **Hold out one fold for testing**

Hold out one of the k folds and designate it as the testing set

- **Train model on remaining folds**

Train the machine learning model on the remaining k-1 folds of data

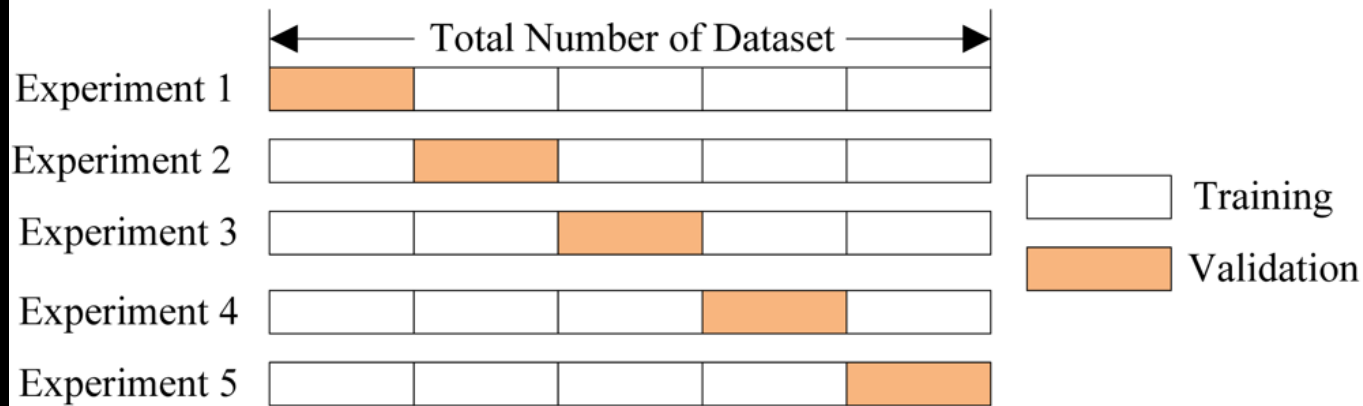
- **Evaluate model on held out fold**

Evaluate the trained model on the held out fold to estimate model performance

- **Repeat for each fold**

Repeat the process by designating a different fold for testing each time and training on the remaining folds

Cross Validation in Credit Models



Cross validation is a technique used to evaluate the performance of a machine learning model on a dataset. In credit models, we can use k-fold cross validation to estimate the performance of a credit scoring model on new, unseen credit applications. The dataset is split into k folds, and the model is trained on k-1 folds and tested on the remaining fold. This is repeated until each fold has been used for testing. The performance over all k folds gives an estimate of how the model will perform on new data.

Understanding

Bias - Variance Tradeoff

MISSING INFO

